

GE Healthcare

Signa HD LPCA to HDx LPCA Upgrade

Technical Publication



5177004
Rev 4

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

© 2006 – 2007, General Electric Company
GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3200 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



Table of Contents

DESCRIPTION	PAGE
Table Of Contents	1
Language Policy For Service Documentation	3
Revision History	5
SECTION 1 – Overview	7
SECTION 2 – Upgrade Installation Process	13
SECTION 3 – Power Off	17
SECTION 4 – Upgrade Procedure	18
SECTION 5 – Completion	36

This page left blank intentionally.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
4-27-2006	1	Initial Release
5-11-2007	2	Section 4: Updated the 1.5T and 3T cable maps with information on Reed Switch and RF cable connections. Added instructions on transferring Reed Switch to new Rear Pedestal (CQA 13100305). Refined instructions on connecting RF cable to Rear Pedestal (CQA 13100313). Removed Isolation Boxes for TwinSpeed upgrade (CQA 13100312). Updated Rear Pedestal Covers information and installation procedures.
10-11-2007	3	Section 4: Updated Subsection 4-4 by adding step to install right-angle connector to J4 so Run 1328 is easier to route and connect. (PQR 13100319)
26-Oct-2010		Section 1: Added required MR training notice and 200 gauss plot. Sections 1 & 2: Revised Ferrous Material Handling warning, Section 4: Revised Ferrous Material Handling warnings and Warning instructions for moving rear pedestal. (CAPA 1859293)

List Of Effective Pages

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title Page		4							
1 to 37		4							

This page left blank intentionally.

SECTION 1 – OVERVIEW

1-1 INTRODUCTION

This manual provides instructions to upgrade a system with an HD LPCA to HDx LPCA for 1.5T or 3.0T.

This Direction covers the following catalogs:

M3335PN – 3.0T HD LPCA to HDx LPCA Upgrade

M3335PZ – 1.5T HD LPCA to HDx LPCA Upgrade

This upgrade includes:

- New Rear Pedestal with components and Rear Split Bridge attached
- New HDx LPCA
- HDx Rear Pedestal covers (for existing HD systems)

WARNING!

FERROUS MATERIAL HAZARD!

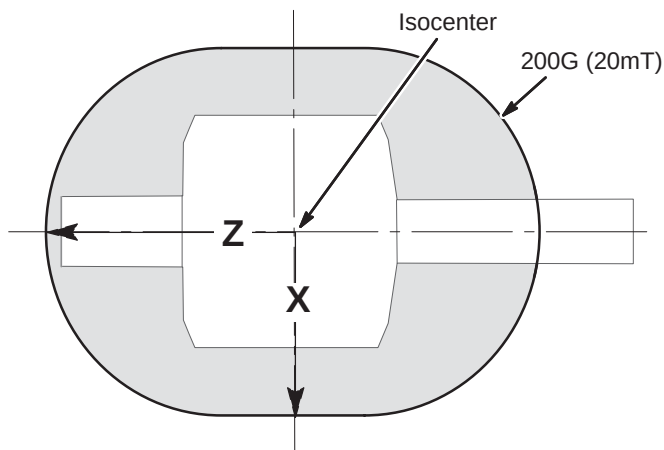
DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.

Illustration is not to scale



Approximate Distance from isocenter to 200G Line		
	X	Z
3.0T	180 cm (71 in) ¹	272 cm (107 in) ¹
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
1.0T & Lower	145 cm (57 in) ¹	195 cm (77 in) ¹
Unshielded Magnet	Measure ²	

Note 1:
These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed.

Note 2:
For unshielded magnets, the magnetic field must be measured using a gauss meter.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of the box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-6 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-6-1 MATERIAL POLICY OVERVIEW

1-6-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-6-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-6-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment de-installed for the HDx upgrade to GoldSeal.

1-6-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new components were shipped to site for return of above material.
- Packing material

1-6-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for any deinstallation procedures of the site.

1-6-5 REGION SPECIFIC RETURN INFORMATION

1-6-5-1 Sites in Americas

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997
Fax: (414)-747-6855
Field ASPEN: 1-800-525-1516, Box #26041

1-6-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-6-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41
Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-6-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-8.

1-7 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page – http://supportcentral.ge.com/15563](http://supportcentral.ge.com/15563)

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-8 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

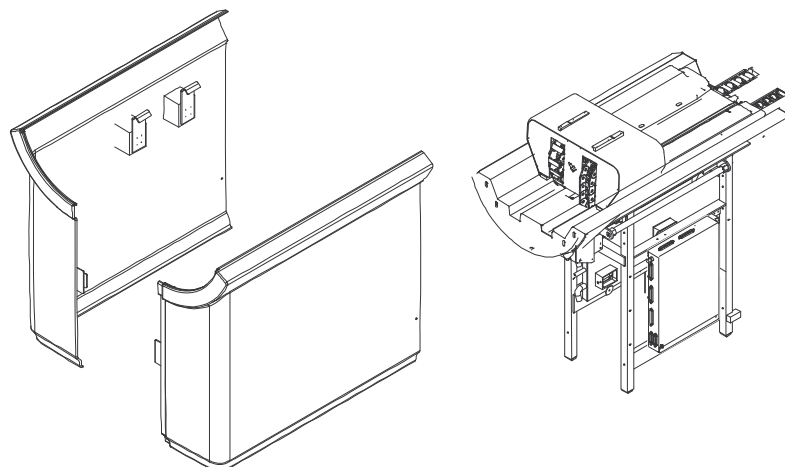
Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Confirm back-up site information. Shut site down.
- Remove any obsolete equipment and cables.
- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software. (If applicable)
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-9 UPGRADE SYSTEM COMPONENTS

Illustration 1-1 shows the major equipment modified for this upgrade.



SIGNA HARDWARE FOR UPGRADE TO HDx LPCA CONFIGURATION
ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading a Signa HD LPCA to an HDx LPCA.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered. **Make sure that every part required is available before starting.**

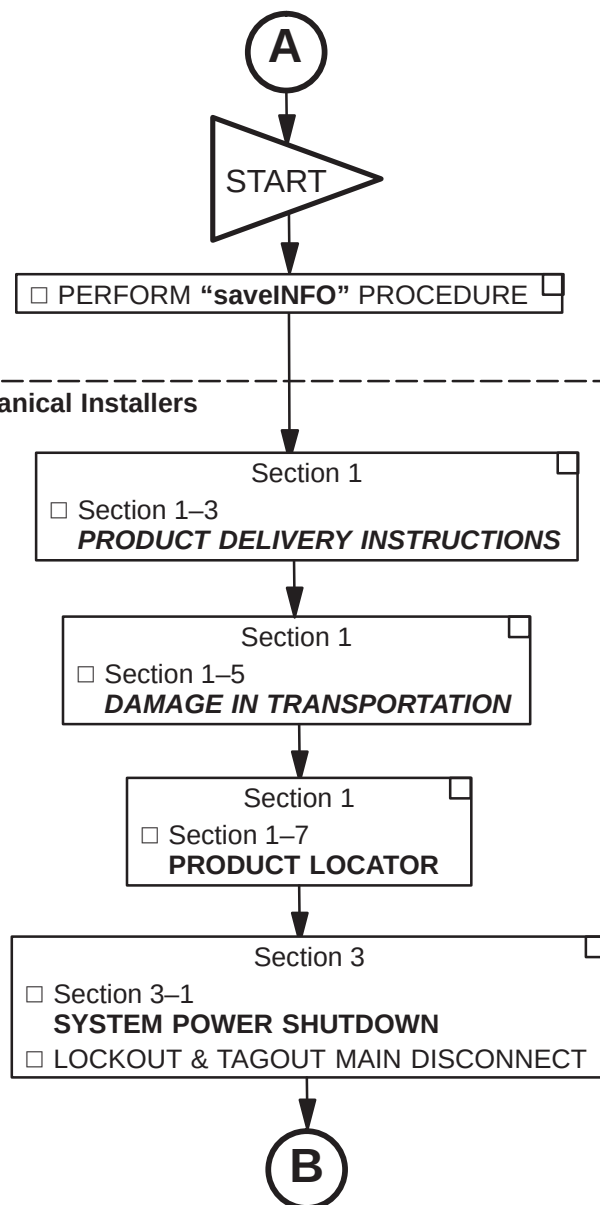
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by Field Engineer



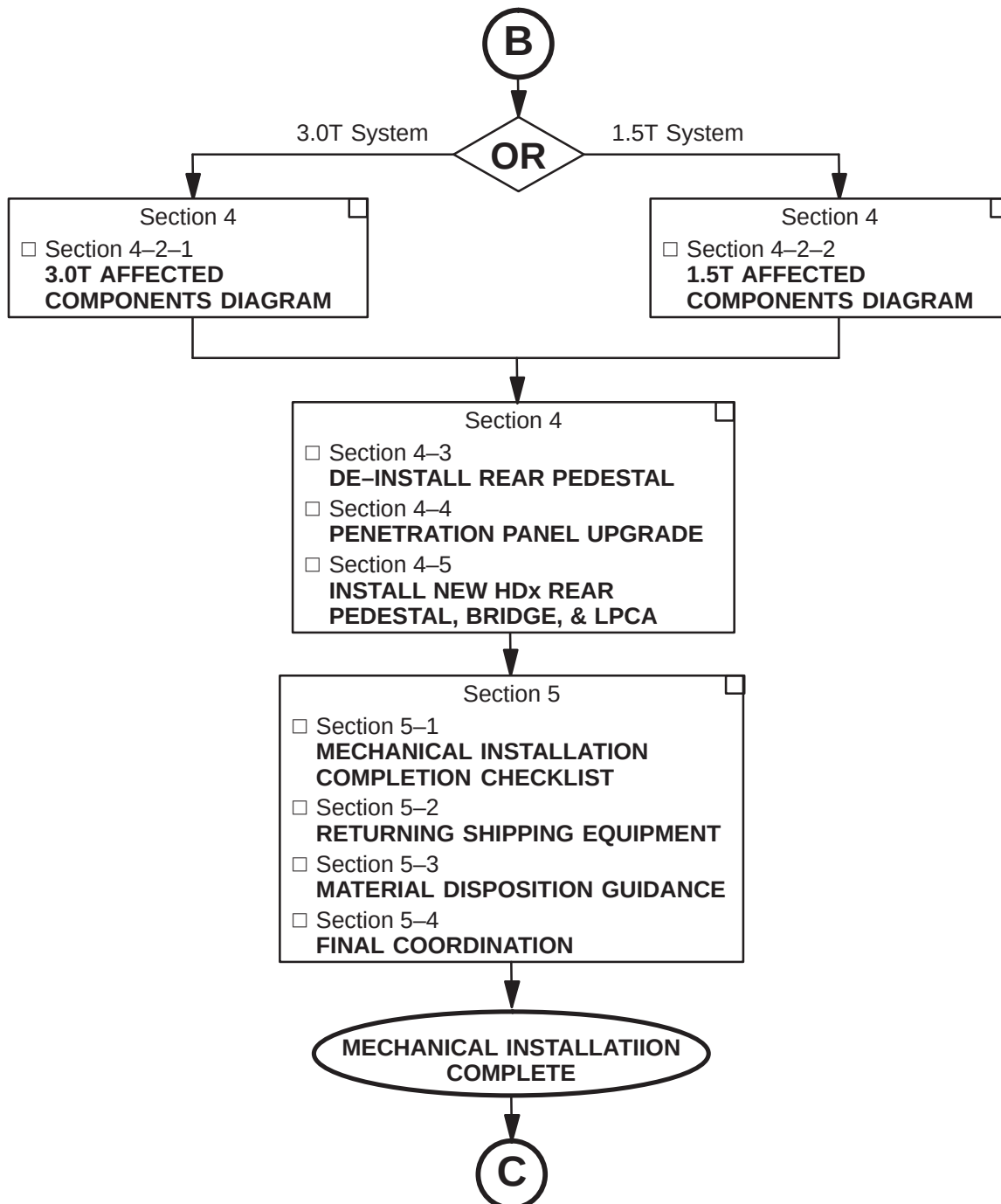
IMPORTANT : PER ISO
9001 REQUIREMENTS

Procedures to be Completed by FE and/or Mechanical Installers



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

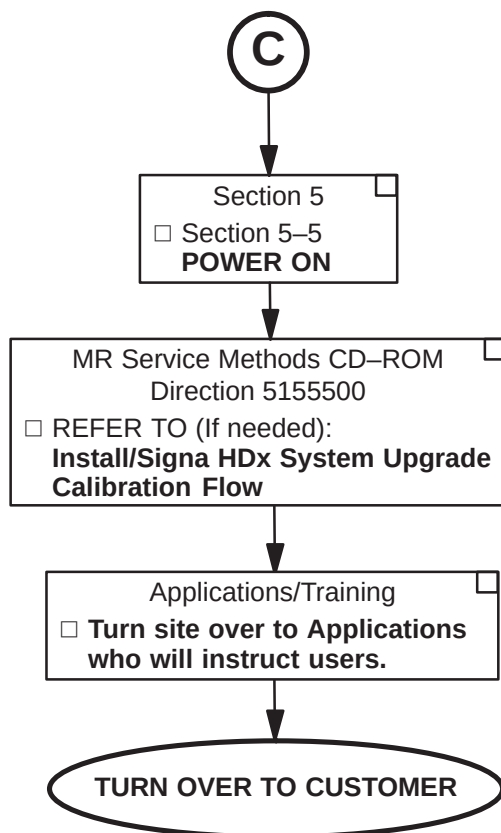


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and Applications



IMPORTANT : PER ISO
9001 REQUIREMENTS



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the POWER OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – UPGRADE PROCEDURE

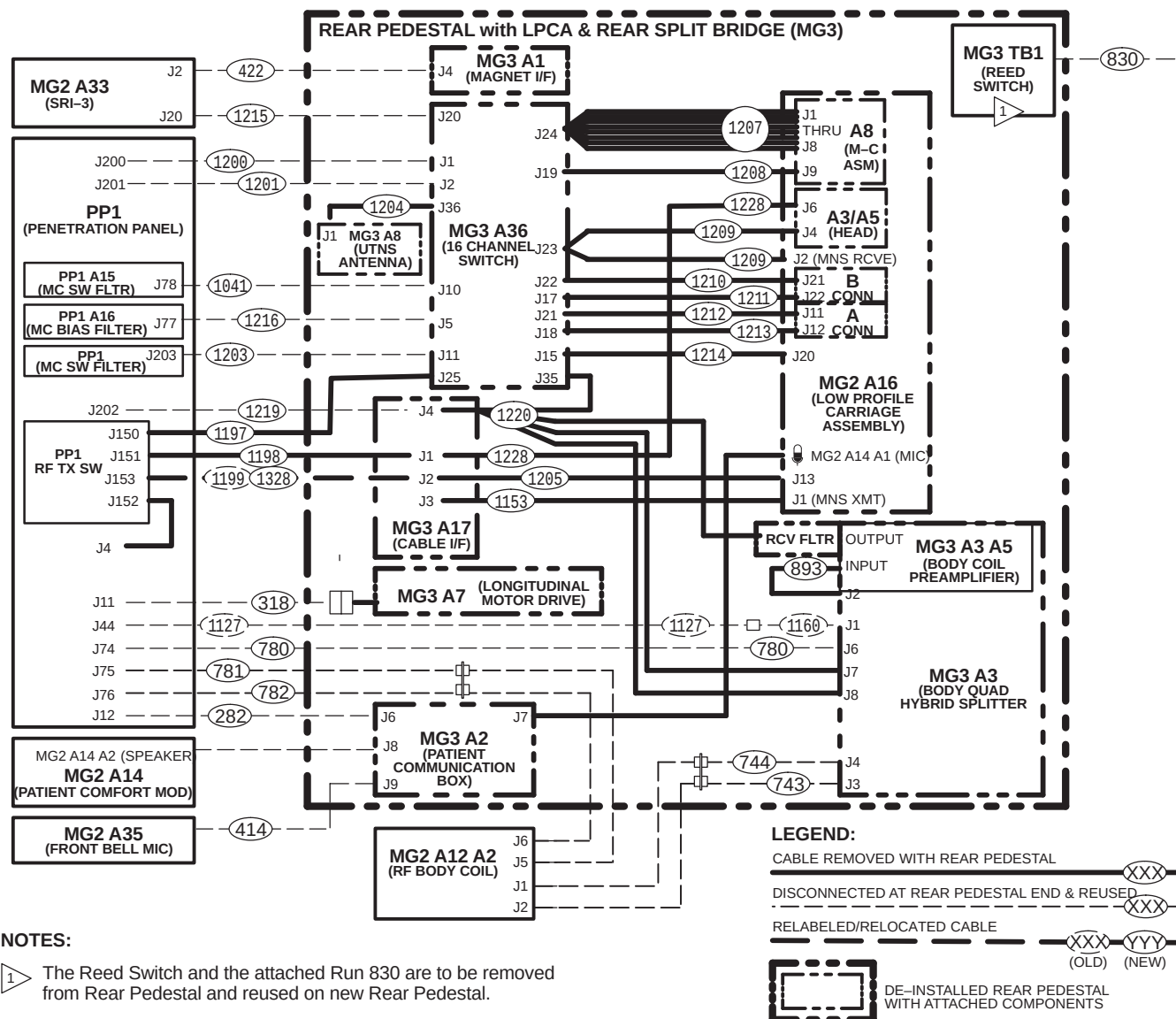
4-1 OVERVIEW FOR 3.0T SYSTEM

The following procedures provide information for upgrading the Rear Pedestal of a 1.5T or 3.0T system that was previously upgraded to HDx (14.x) from EXCITE (11.x). This EXCITE to HDx type of upgrade retained the EXCITE LPCA which needs to be updated to the new style HDx LPCA. This is required for a 32 Channel system.

4-2 EXISTING REAR PEDESTAL COMPONENTS AFFECTED BY UPGRADE

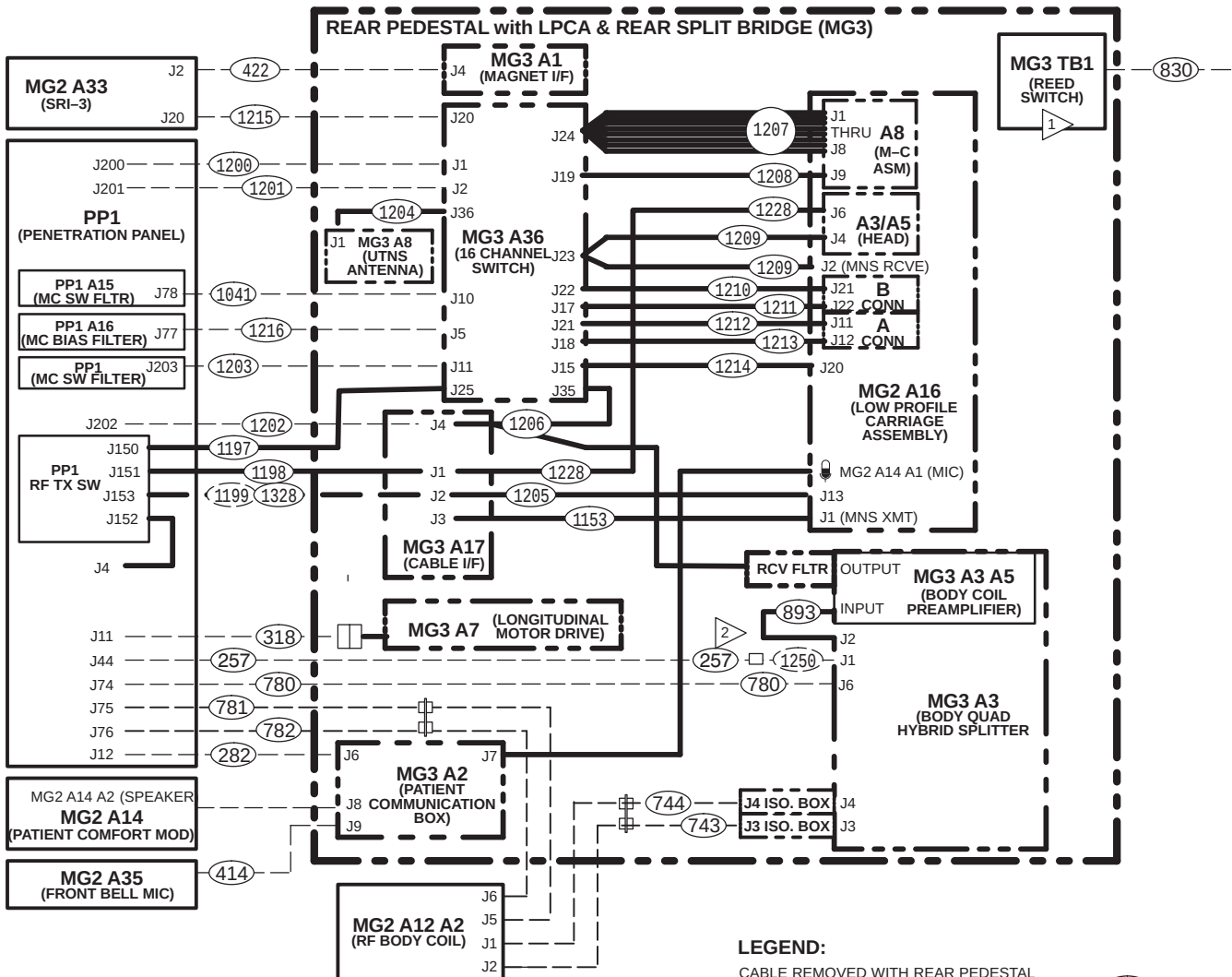
4-2-1 3.0T SYSTEM

The illustration below indicates the cables to be de-installed, relabeled, and reused along with the Rear Pedestal and attached components that will be de-installed for a 3.0T system.



4-2-2 1.5T SYSTEM

The illustration below indicates the cables to be de-installed, relabeled, and reused along with the Rear Pedestal and attached components that will be de-installed for a 1.5T system.



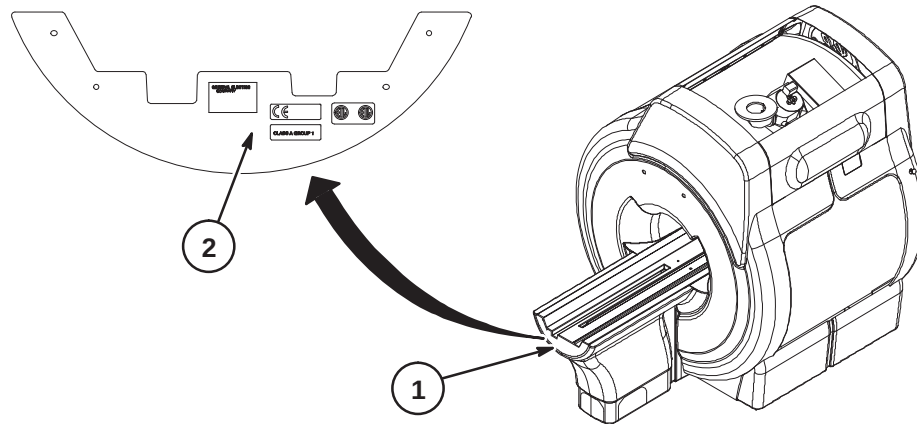
NOTES:

- 1 The Reed Switch and the attached Run 830 removed from the de-installed Rear Pedestal and attached to the new Rear Pedestal.
- 2 Systems upgraded to Signa HD may not have the RF flexible extension cable, Run 1250, installed. The existing RF cable, Run 257, has a right-angle connector that connects directly to the Hybrid Splitter.

4-3 DE-INSTALL REAR PEDESTAL

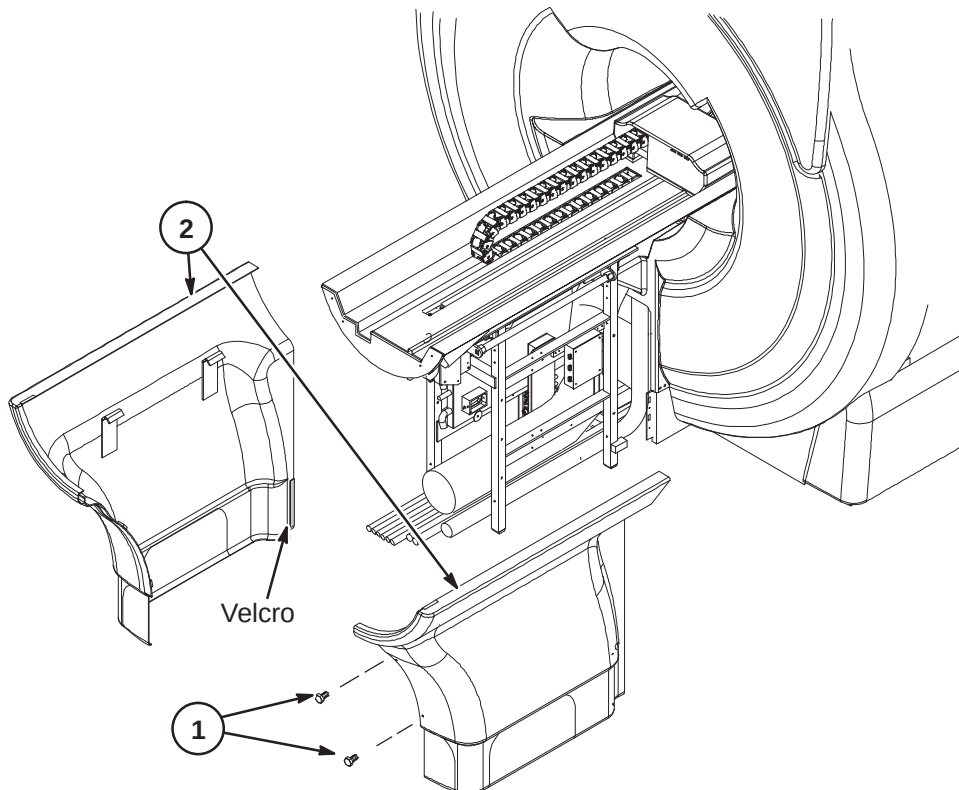
4-3-1 UNFASTEN REAR BRIDGE PLATE

1. Remove existing Rear Bridge Plate.
2. Save for re-attaching plate to new rear split bridge. Any new rating plates will be attached to this plate.



4-3-2 DE-INSTALL REAR PEDESTAL COVERS

1. Unfasten screws holding covers together.
 2. Lift covers off side bar and return for recycling.
- Note:** Covers are also held in place with Velcro on lower part of covers as shown below.



4-3-3 DISCONNECT CABLES FROM HYBRID SPLITTER

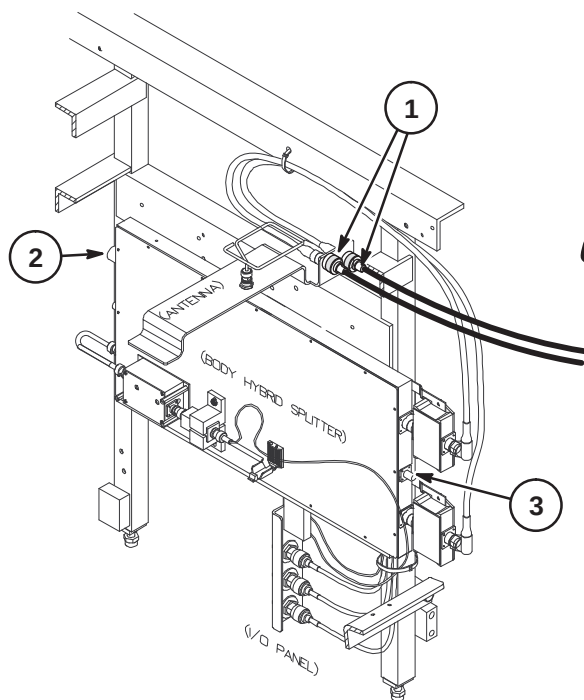
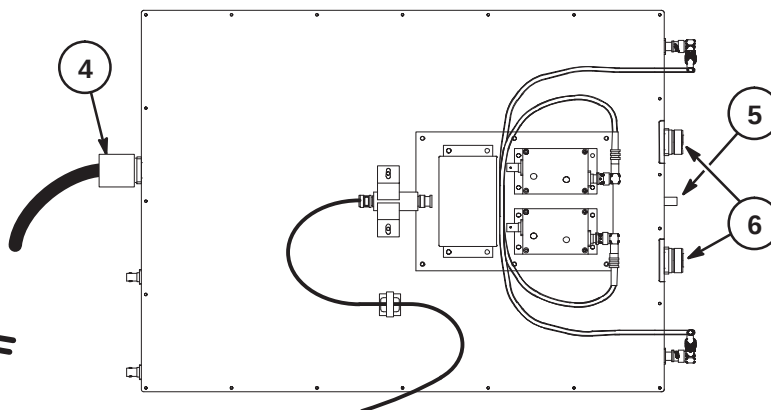
The RF Drive Input cables utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

FOR 1.5T HYBRID SPLITTER

1. Disconnect and save I and Q RF Drive input cables from RF Coil from J1 and J2 on bracket.
2. Disconnect cable connected to J1. On early systems, Run 257 was equipped with a right-angle connector and was connected directly to J1. On later systems, RF Flexible cable Run 1250 was connected to J1. The other end of Run 1250 was connected to Run 257.
3. Disconnect and save Run 780 from J74 on Penetration panel to J6 at front of Hybrid Splitter.

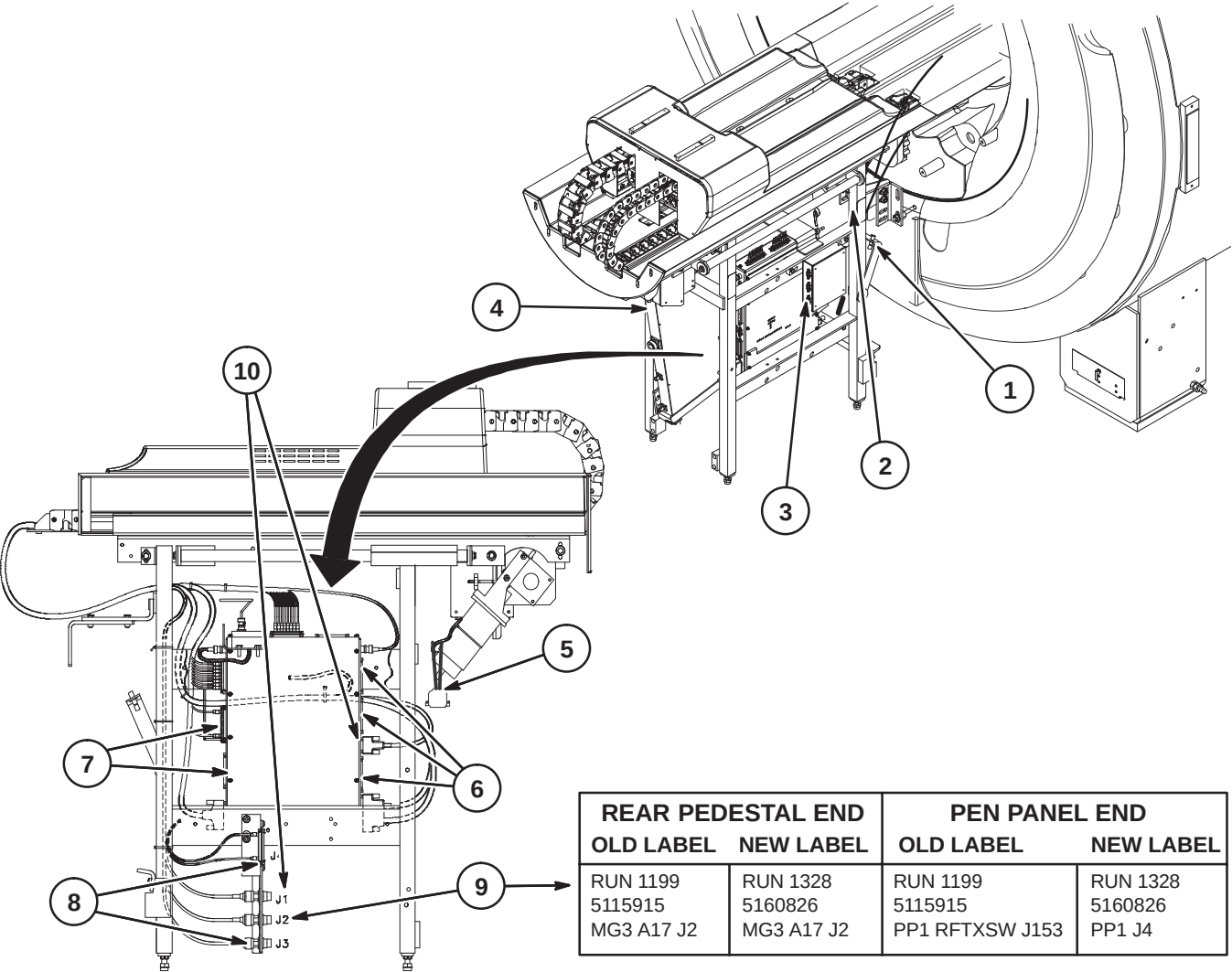
FOR 3.0T HYBRID SPLITTER

4. Disconnect and save RF Flexible cable Run 1160 from J1. Other end is connected to Run 1127.
5. Disconnect and save Run 780 from J6 at front of Hybrid Splitter.
6. Disconnect and save I and Q RF Drive input cables from RF Coil from J3 and J4 on Splitter.

**1.5T HYBRID SPLITTER****3.0T HYBRID SPLITTER**

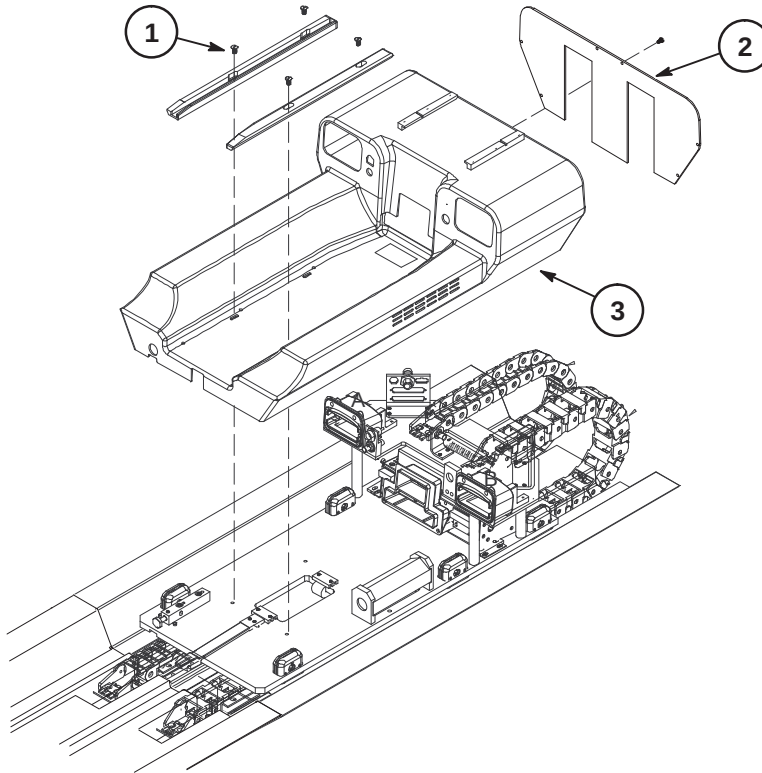
4-3-4 DISCONNECT REMAINING CABLES FROM REAR PEDESTAL

- 1. Disconnect Runs 781 and 782 Bias cable from J5 and J6 on bracket.
 - 2. The Reed Switch, usually attached to the leg of the Rear Pedestal and conneted to Run 830, must be unfastened and retained for installation in new Rear Pedestal.
 - 3. On the Patient Communication Box, disconnect Run 282 to J6, the Speaker cable from J8, and the Front End Bell Microphone cable from J9.
 - 4. Disconnect Run 422 from J4 on Magnet Interface .
 - 5. Disconnect Run 318 from the Longitudinal Drive Motor cable.
 - 6. On the 16 Channel Switch, disconnect Run 1215 from J20 and Run 1216 from J5.
 - 7. On the 16 Channel Switch, disconnect Run 1200 from J1, Run 1201 from J2, Run 1203 from J4, and Run 1041 from J10.
 - 8. On the Bulkhead Interface, disconnect Run 1219 from J4 and, if MNS option is installed, Run 1147 from J3.
- Note:** Runs 1197, 1198, and 1199 referred to in the following two steps are found only on a HDx (14.x) system that was upgraded from an EXCITE (11.x) or HD (12.x) system.
- 9. Disconnect Run 1199 from J2. This cable will be relabeled as Run 1328. Refer to table below.
 - 10. Disconnect Runs 1197 from J25 on 16 Channel Switch and Run1198 from J1 on Bulkhead I/F. These cables will be de-installed and returned for recycling. Other end of cable is connected to Penetration Panel.

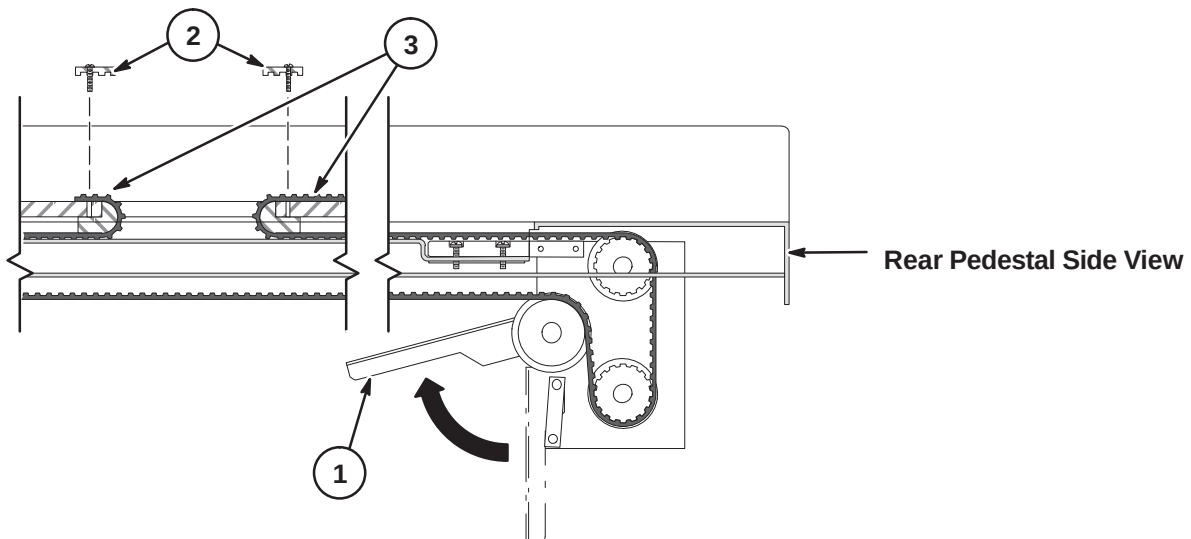


4-3-5 REMOVE HD LPCA COVER

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover from LPCA
3. Carefully remove LPCA cover from base plate assembly.

**4-3-6 DISCONNECT DRIVE BELT**

1. Release tension arm.
2. Unfasten clamp screws and remove clamps.
3. De-install belt from Rear Pedestal and Rear Split Bridge. Do not unwind it from Front Split Bridge pulley. Save for installation in new HDx Rear pedestal assembly.

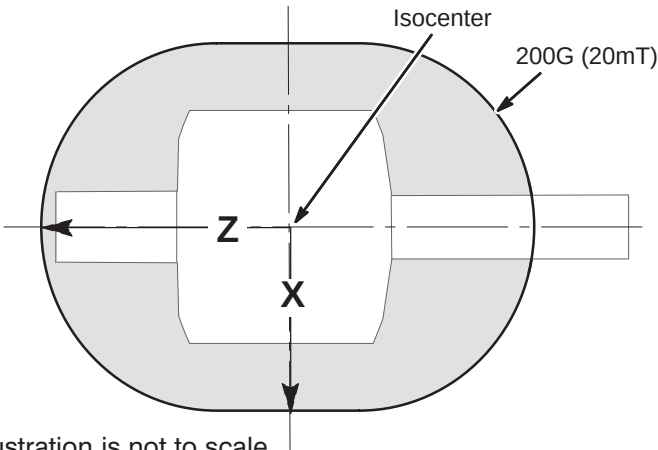


4-3-7 200 GAUSS ZONE LIMIT FOR MOVEMENT OF FERROUS COMPONENTS

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.



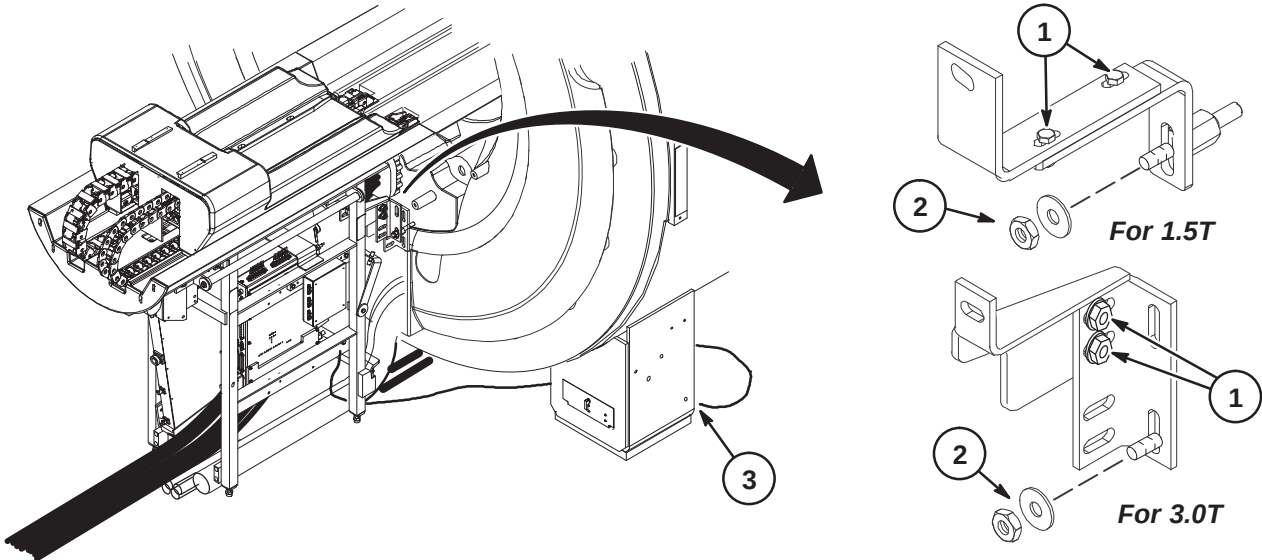
THE LONGITUDINAL DRIVE MOTOR IN THE REAR PEDESTAL IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE DRIVE MOTOR MUST BE OUTSIDE THE 200 GAUSS ZONE.



Approximate Distance from isocenter to 200G Line		
	X	Z
3.0T	180 cm (71 in) ¹	272 cm (107 in) ¹
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

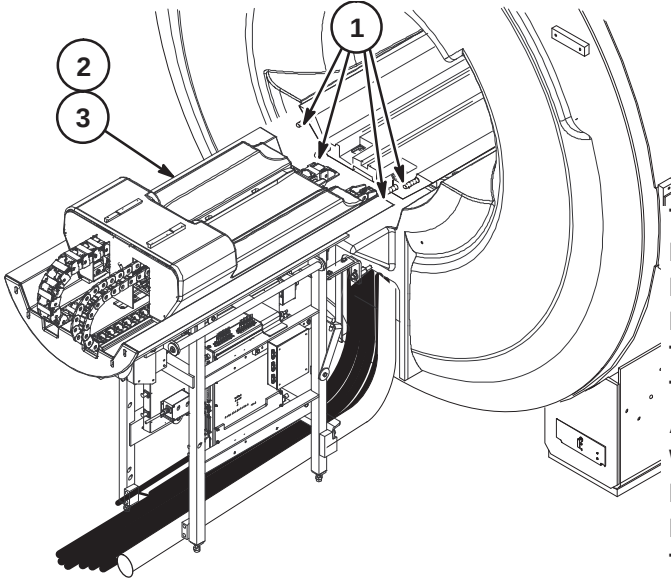
4-3-8 UNFASTEN REAR PEDESTAL FROM MAGNET FOR 1.5T AND 3.0T

1. Loosen, do not remove, fasteners holding brackets together.
2. Remove M10 Nut and Washer from each stud. Save for installing new HDx Rear Pedestal assembly.
3. Disconnect Rear Pedestal ground wire from ground stud on magnet leg.



4-3-9 UNFASTEN REAR SPLIT BRIDGE FROM FRONT SPLIT BRIDGE

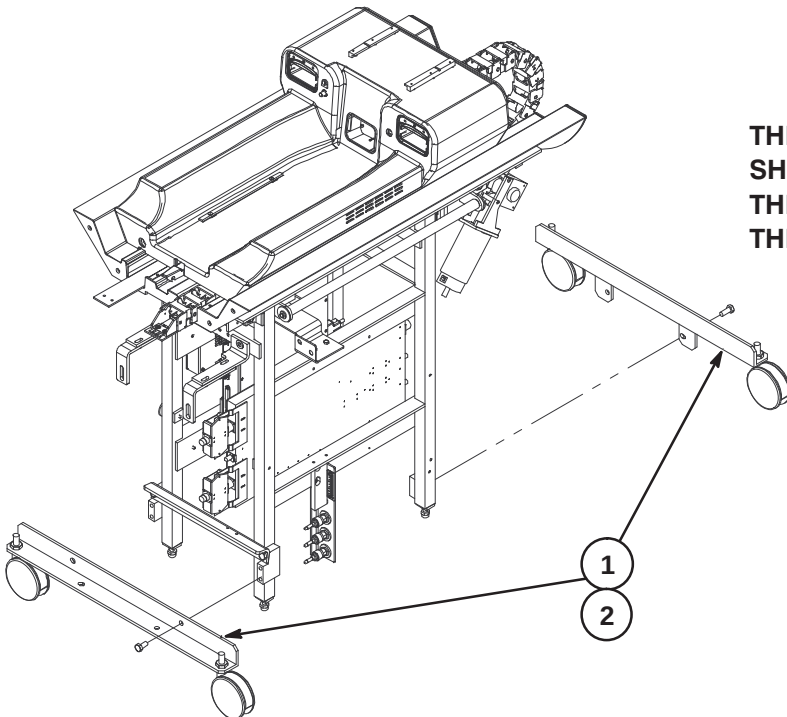
1. Remove four M10 hex nuts (2109877-3) and washers (2228717-2) fastening the Front and Rear Split Bridge pieces together. These will be reused for attaching the new Rear Split Bridge to the existing Front Split Bridge.
2. Using three or more people, carefully move Rear Pedestal with Bridge and LPCA away from magnet.
3. Using three or more people, move the Rear Pedestal assembly out of magnet Room.

**WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! THE REAR PEDESTAL CAN WEIGH UP TO 310 LBS. (132 KG). IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE. REFER TO SECTION 4-3-7. WHEN REMOVING REAR PEDESTAL, IT MUST BE MOVED DIRECTLY BACK FROM THE MAGNET AND NOT ROTATED. USE PROPER SAFETY AND LIFTING TECHNIQUES.

4-3-10 RETURN HD REAR PEDESTAL ASSEMBLY FOR RECYCLING

1. Attach Front and Rear shipping brackets to removed HD Rear Pedestal. These brackets were used to ship the new HDx Rear Pedestal to site.
2. Mount and fasten to shipping pallet used for bringing new HDx Rear Pedestal to site.

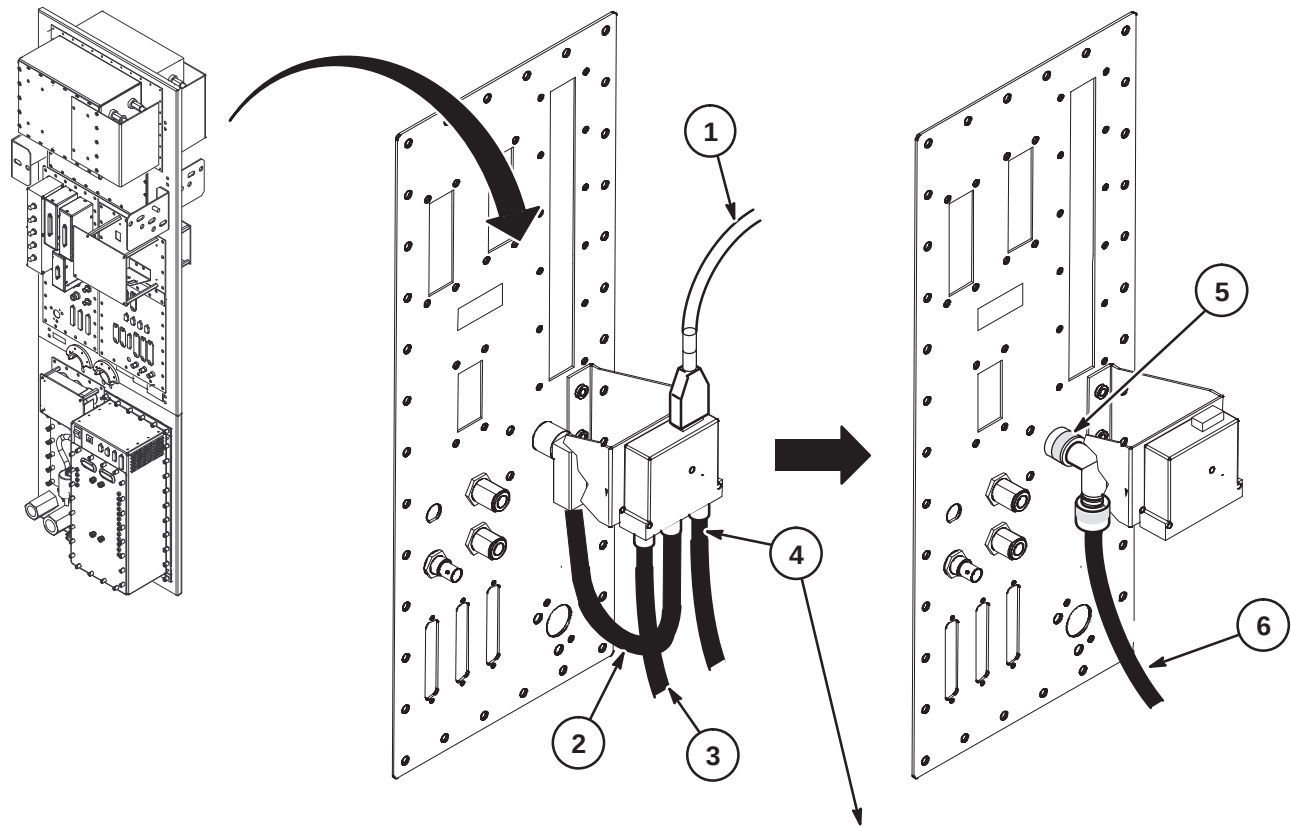
**WARNING!**

THE REAR PEDESTAL FRONT AND REAR SHIPPING BRACKETS ARE FERROUS! THEY MUST NOT BE BROUGHT INTO THE MAGNET ROOM.

4-4 PENETRATION PANEL UPGRADE

If site being upgraded has an EXCITE (11.x) or HD (12.x) Penetration Panel, then the RF Transmit Switch on the Magnet Room side of the Penetration Panel will not be used for the 32 Channel upgrade.

- 1. Disconnect Run 1197 from J150 and return for recycling.
- 2. Disconnect short flexible cables between J4 and J152 and return for recycling.
- 3. Disconnect Run 1198 from J151 and return for recycling.
- 4. Disconnect Run 1199 from J153 and relabel to Run 1328 as shown in table below.
- 5. Install 46-221865P3 right-angle connector on J4 on Penetration Panel. Orientation specific to site needs.
- 6. Connect relabeled Run 1328 to J4 on Penetration Panel.



REAR PEDESTAL END		PEN PANEL END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN 1199 5115915 MG3 A17 J2	RUN 1328 5160826 MG3 A17 J2	RUN 1199 5115915 PP1 RFTXSW J153	RUN 1328 5160826 PP1 J4

4-5 INSTALL NEW HDx REAR PEDESTAL, BRIDGE, AND LPCA

Follow instructions for unpacking new HDx Rear Pedestal before moving into magnet room.

WARNING!

FERROUS MATERIAL HAZARD! If magnet is ramped, the wheel assemblies attached to the Rear Pedestal for shipping and delivery MUST be removed before moving Rear Pedestal into Magnet Room.

CAUTION

The Rear Pedestal can weigh up to 310 lbs. (132 kg) and requires at least THREE people to move Rear Pedestal into position at rear of magnet.

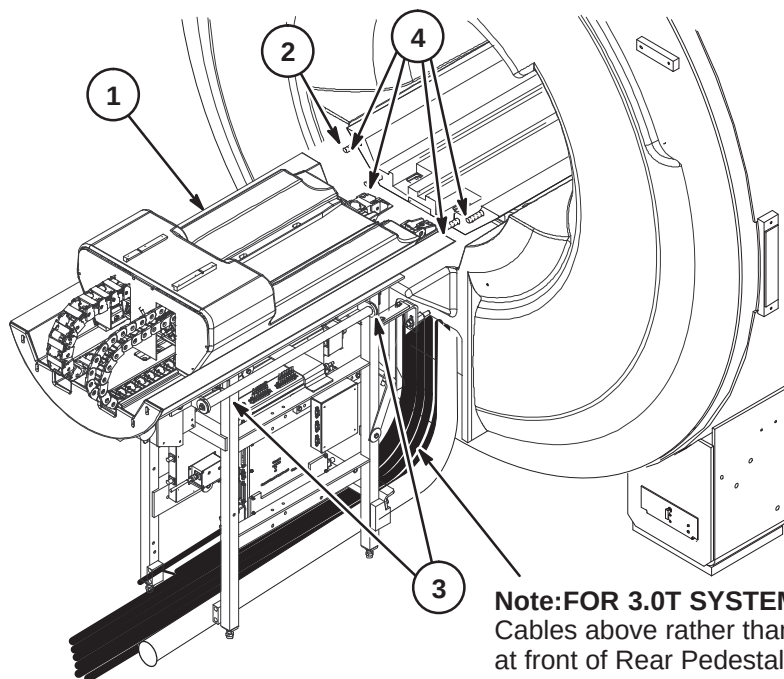
Note: FOR 3.0T SYSTEM – Routing the Gradient Cables above rather than below the cross brace at front of Rear Pedestal will reduce white pixels.

4-5-1 ATTACH REAR PEDESTAL

1. Move Rear Pedestal into position at rear on magnet as shown below. Make sure to follow instructions in warning statement below.

Note: Dress and align all cables and hoses routed thru Rear Pedestal while moving Rear Pedestal closer to magnet.

2. Move Rear Pedestal forward and engage front split bridge studs.
3. Adjust the Rear Pedestal on each side to level the Rear Split Bridge with the Front Split Bridge.
4. Fasten Front and Rear Split Bridge with four nuts and washers that were removed when de-installing old Rear Pedestal.



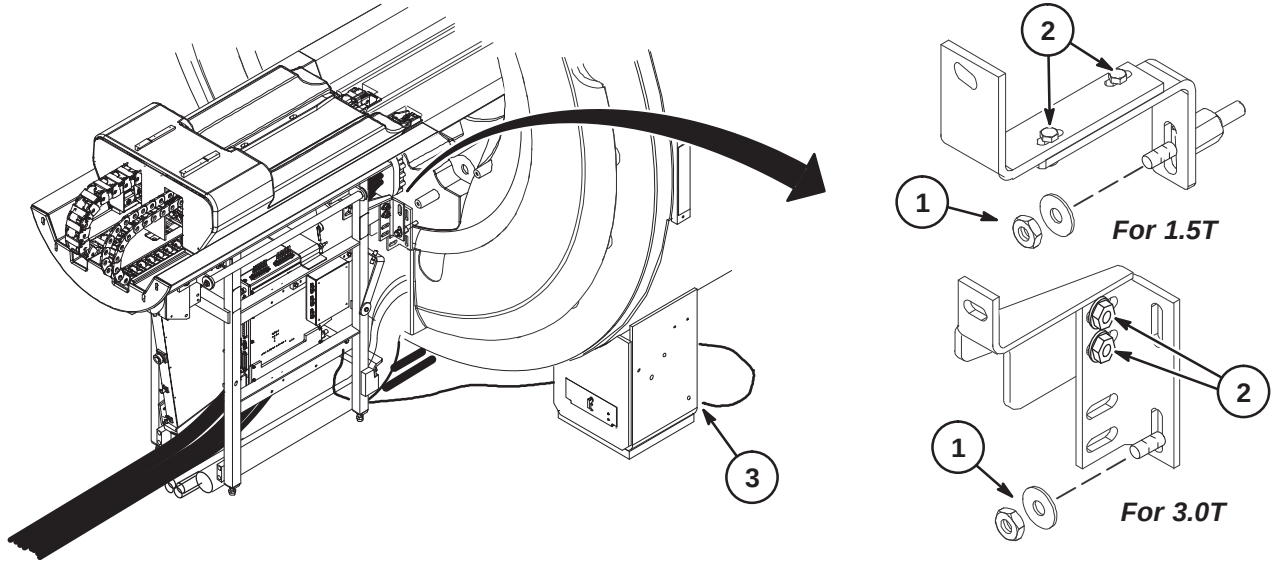
Note:FOR 3.0T SYSTEM – Routing the Gradient Cables above rather than below the cross brace at front of Rear Pedestal will reduce white pixels.

WARNING!

FERROUS MATERIAL HAZARD!
The Rear Pedestal contains ferrous components, namely the Longitudinal Drive Motor. When entering the magnet room with rear pedestal, position the longitudinal drive motor as far away from the magnet as possible.
REFER TO SECTION 4-3-7.

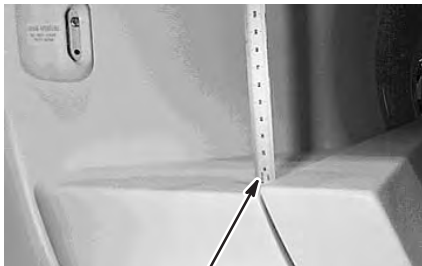
4-5-2 FASTEN REAR PEDESTAL TO MAGNET FOR 1.5T AND 3.0T

1. Center Rear Pedestal on Rear Endbell and attach to Support Studs using M10 Nut (2109875-4) and Flat Washer (2109875-5) on each stud.
2. Adjustment of Rear Pedestal location can be made at this slotted hole location when connecting the Split Bridge.
3. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.

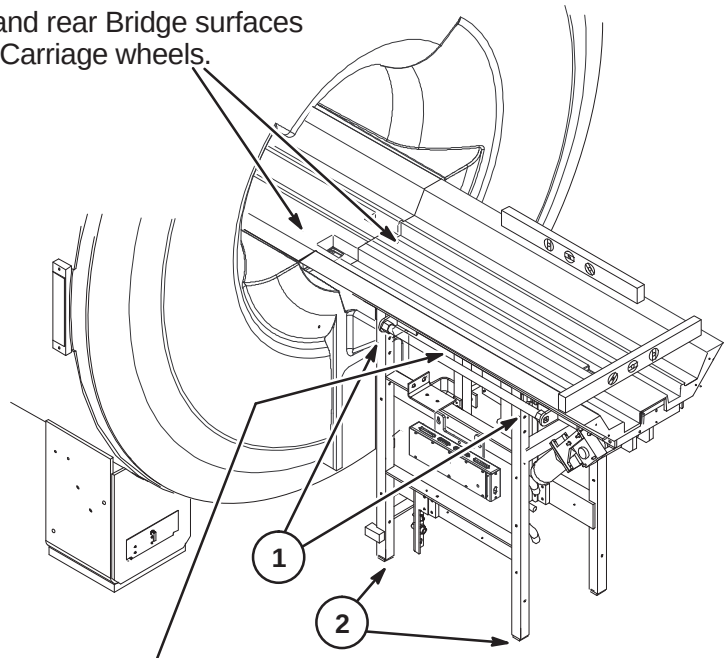
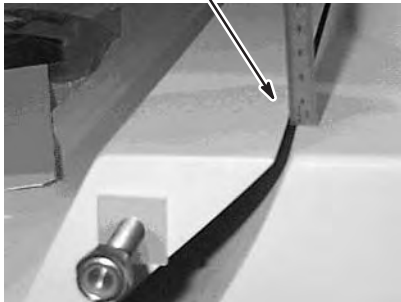
**4-5-3 LEVELING OF BRIDGE**

1. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedestal.

NOTE: Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.



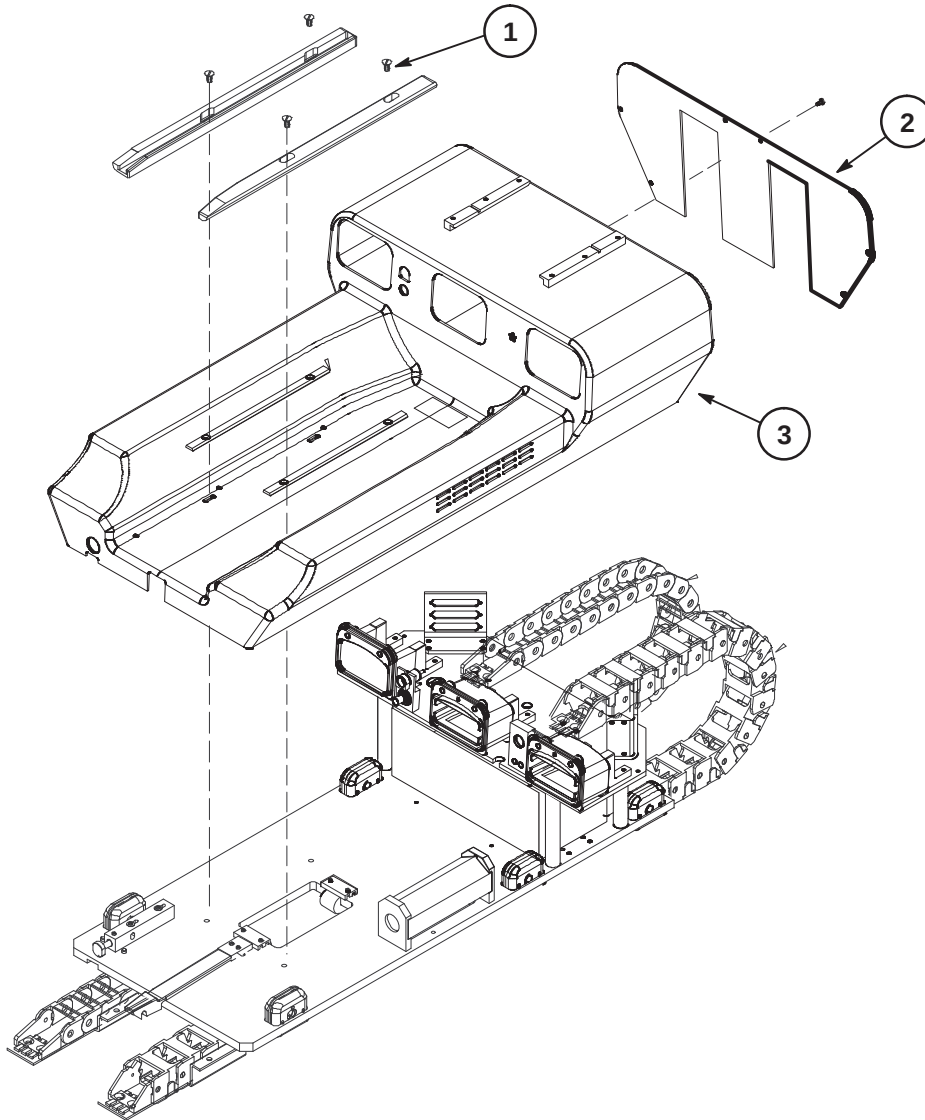
2mm to 4mm above End Bell surface.



NOTE: Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

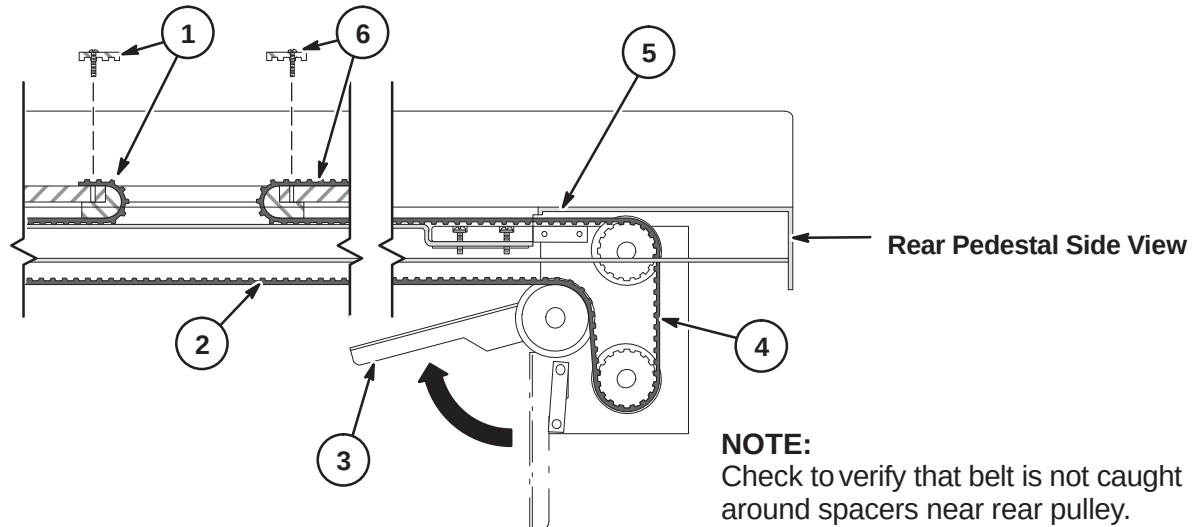
4-5-4 REMOVE HDx LPCA COVER

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover (2386346) from LPCA
3. Carefully remove cover from base plate assembly.

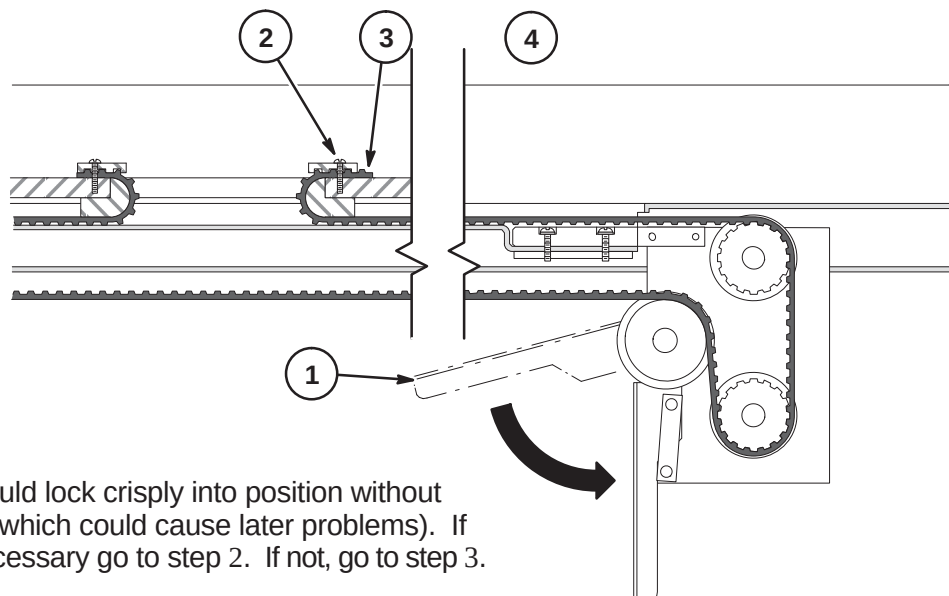


4-5-5 ROUTE AND ATTACH DRIVE BELT

1. Route one end of Drive Belt delivered with Front Split Bridge thru opening in LPCA Base Plate and attach with clamp.
2. Route other end of drive belt from front of Bridge towards Tension Arm Pulley.
3. Release Tension Arm.
4. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
5. Route Belt under and through opening to top of Carriage.
6. Position Belt under Clamp Plate and attach.

**4-5-6 FINAL ADJUSTMENT OF DRIVE BELT**

1. Check Belt tension by closing Tension Arm. (See Note)
2. **(IF NECESSARY):** Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
3. Tighten Clamps and trim excess Belt length.
4. Fasten Carriage Cover to Carriage with four screws removed in Section 4-5-4.



NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

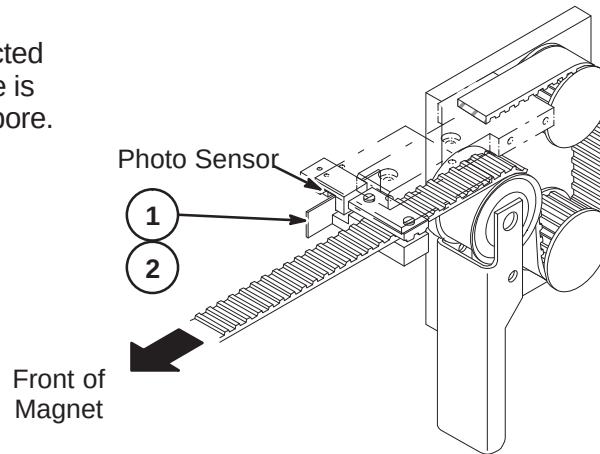
4-5-7 INSTALLATION OF SENSOR FLAG

NOTE: The following Step is needed if flag was removed during routing of drive belt.

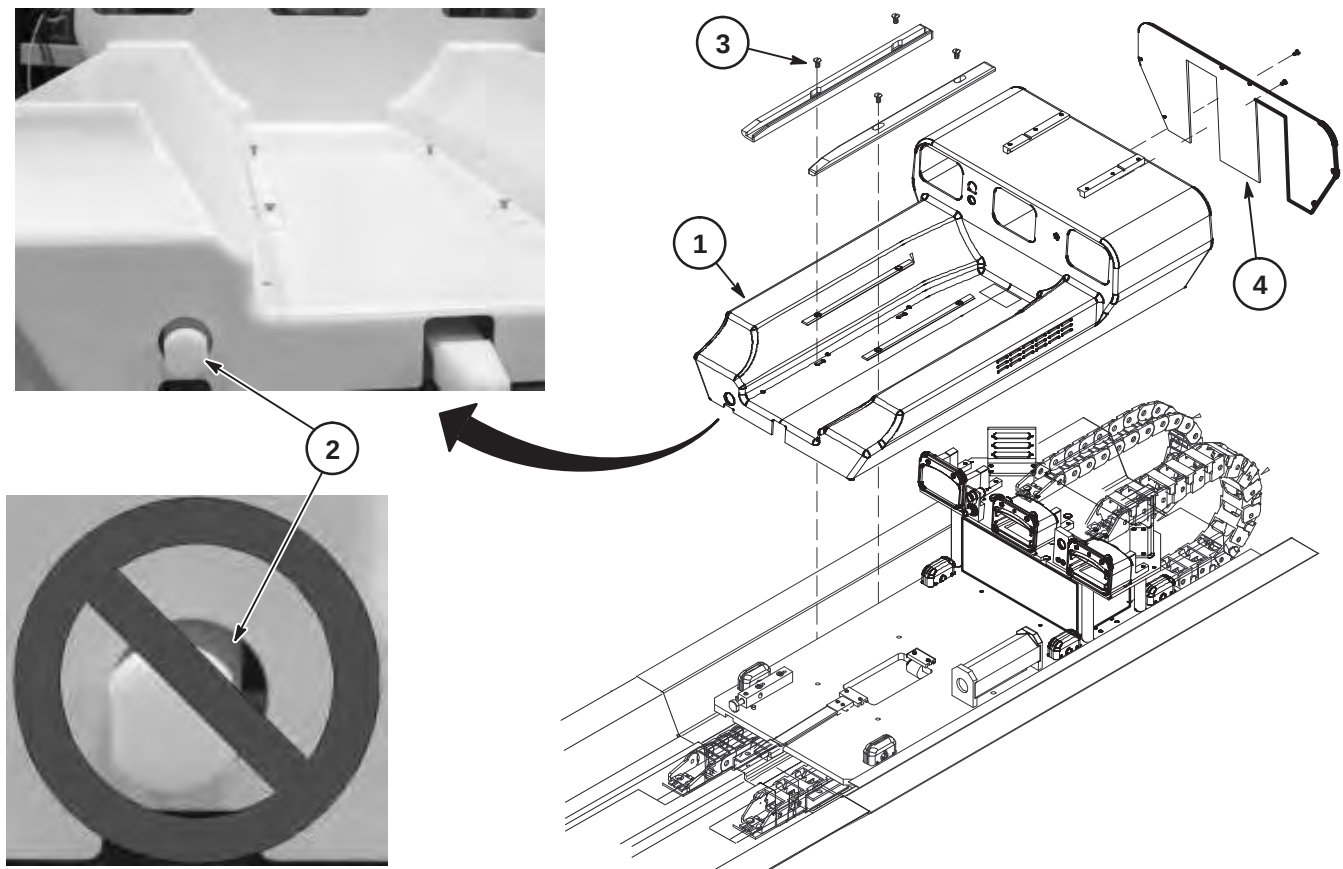
1. Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.
2. Adjust as necessary.

NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

**4-5-8 INSTALLATION OF LPCA COVER**

1. Carefully place cover on base plate assembly.
2. Center the front knob within the hole on the x-axis. The y-axis does not need to be centered.
3. Once centered, secure the cover to the base plate using four screws previously removed.
4. Attach Rear Cover using 6 screws previously removed.



4-5-9 CONNECT CABLES TO HYBRID SPLITTER



The RF Drive Input cables utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

FOR 1.5T HYBRID SPLITTER

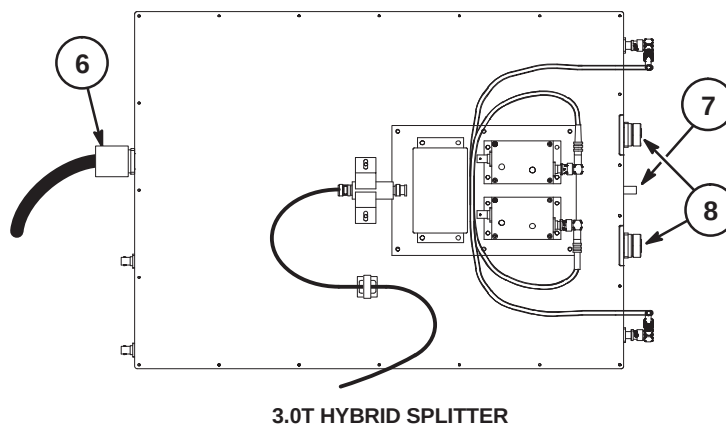
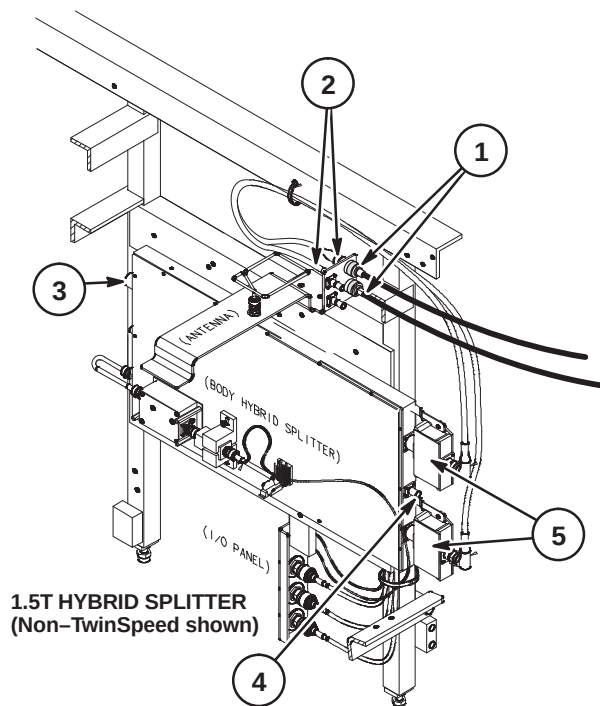
1. Route and connect I and Q RF Drive input cables from RF Coil and connect as marked to J1 and J2 on bracket.

Note: Runs 743 and 744, Body Coil to Hybrid Splitter cable connections, may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

2. Connect Run 743 to J2 and Run 744 to J1 on bracket.
3. If part of system, connect RF Flexible cable Run 1250 to J1. Other end is connected to Run 257. For older systems, Run 257 was assembled with a right-angle connector on the end. Connect this cable directly to J1.
4. Route and connect Run 780 from J74 on Penetration panel to J6 at front of Hybrid Splitter.
5. **FOR 1.5T TwinSpeed ONLY:** Remove the two Isolation Boxes (5118952) pre-installed on Hybrid Splitter and connect I and Q RF Drive cables directly to Hybrid Splitter.

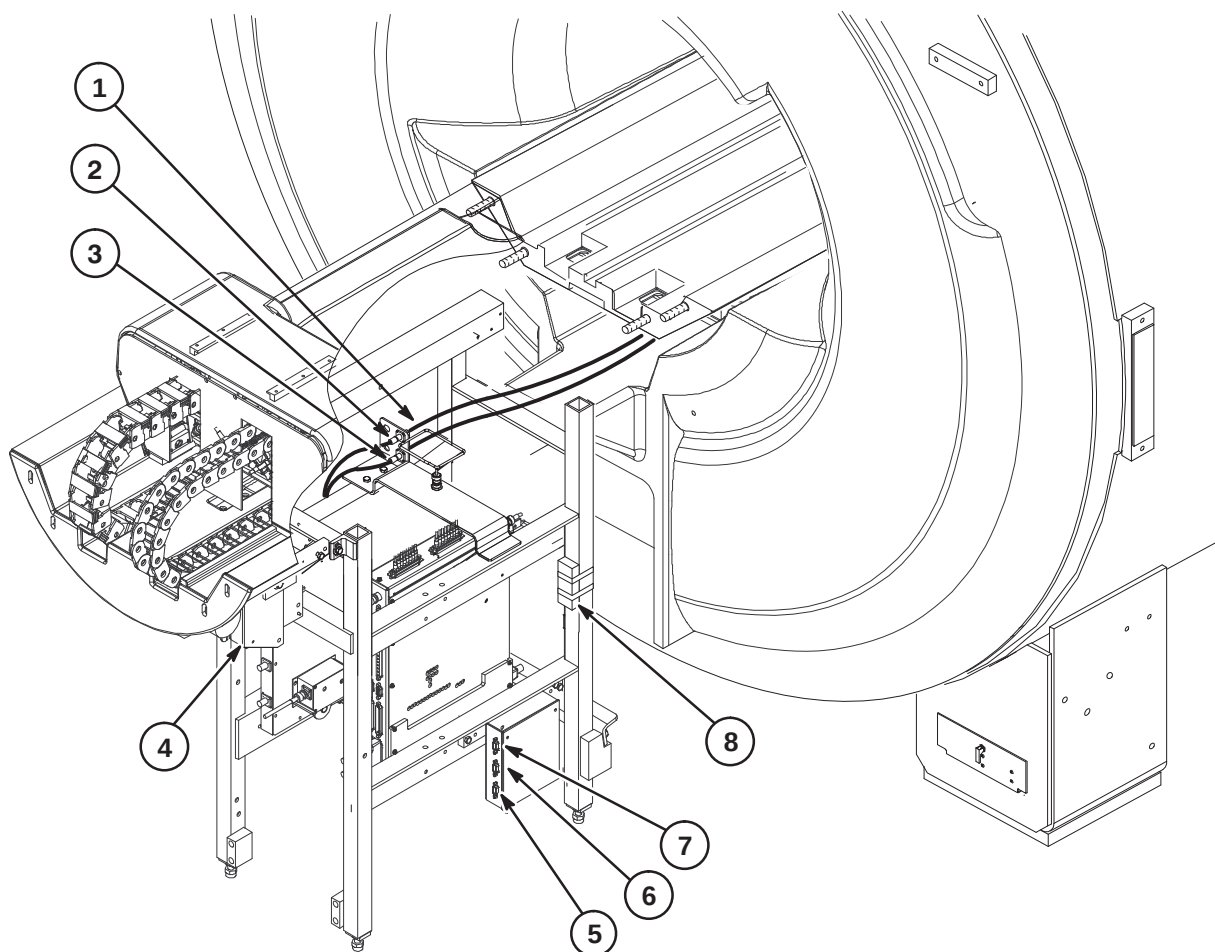
FOR 3.0T HYBRID SPLITTER

6. Connect RF Flexible cable Run 1160 to J1. Other end is connected to Run 1127.
7. Connect Run 780 to J6 at front of Hybrid Splitter.
8. Connect I and Q RF Drive input cables from RF Coil to J3 and J4 on Splitter.



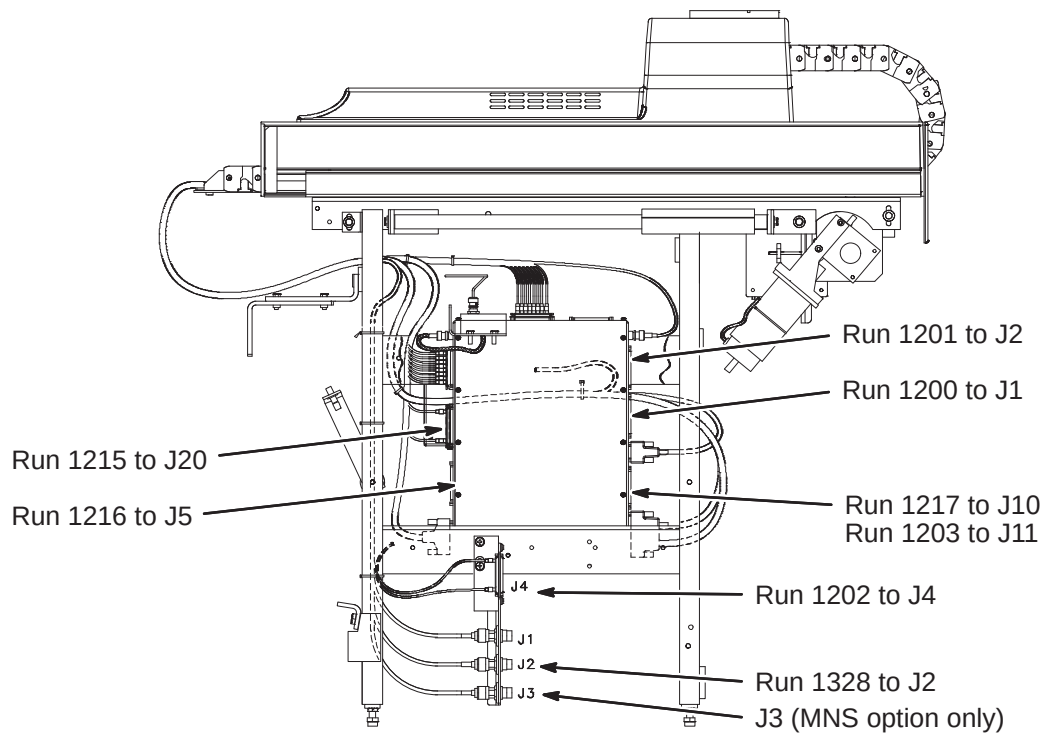
4-5-10 ROUTE AND CONNECT CABLES AT REAR PEDESTAL

1. Route and connect RF Bias cables from RF Coil to side of UTNS bracket.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Patient Comfort Module to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Attach the previously removed Reed Switch to the leg of the Rear Pedestal, Make sure Run 830 is connected to the Reed Switch.

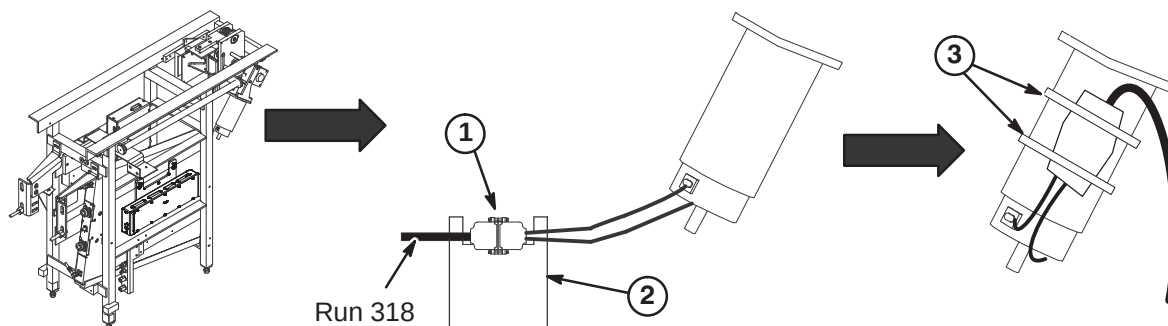


4-5-11 CONNECT CABLES TO 16 CHANNEL SWITCH AND BULKHEAD INTERFACE

Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated below.

**4-5-12 CONNECT RUN 318 TO LONGITUDINAL DRIVE MOTOR**

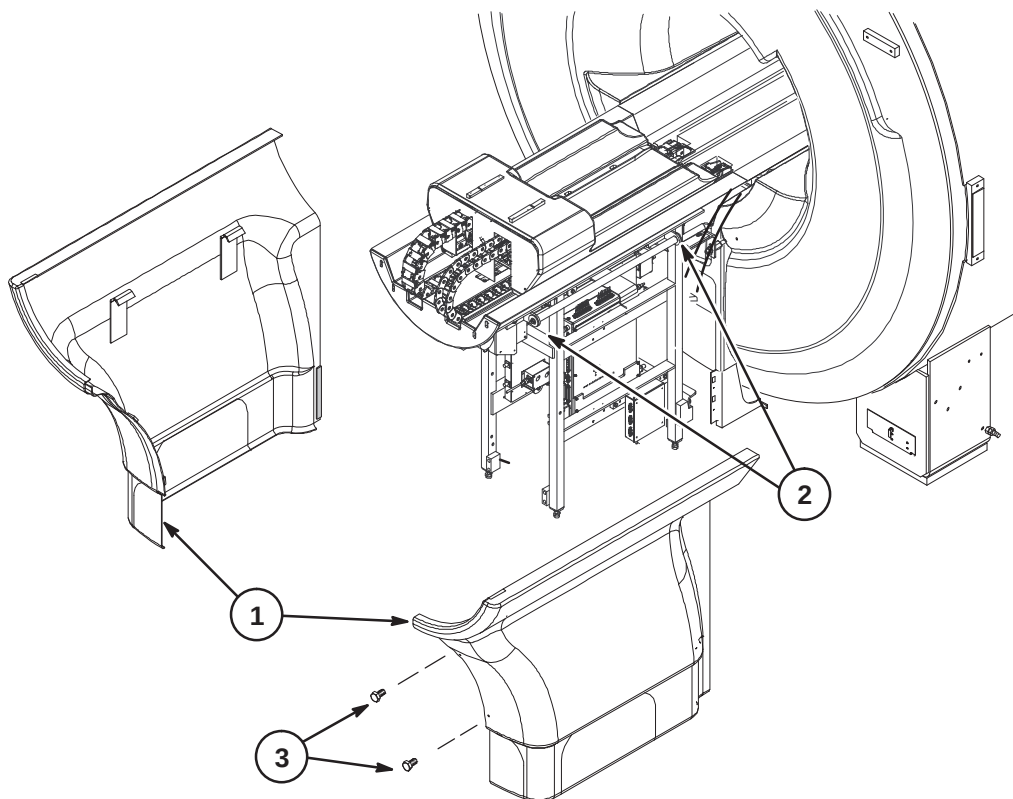
1. Connect Run 318 to Longitudinal Drive Motor cable.
2. Wrap both connectors with 177 x 200 Connector Wrap (2236130-2).
3. Ty-wrap cable to motor to prevent movement.



4-5-13 INSTALL REAR PEDESTAL COVERS

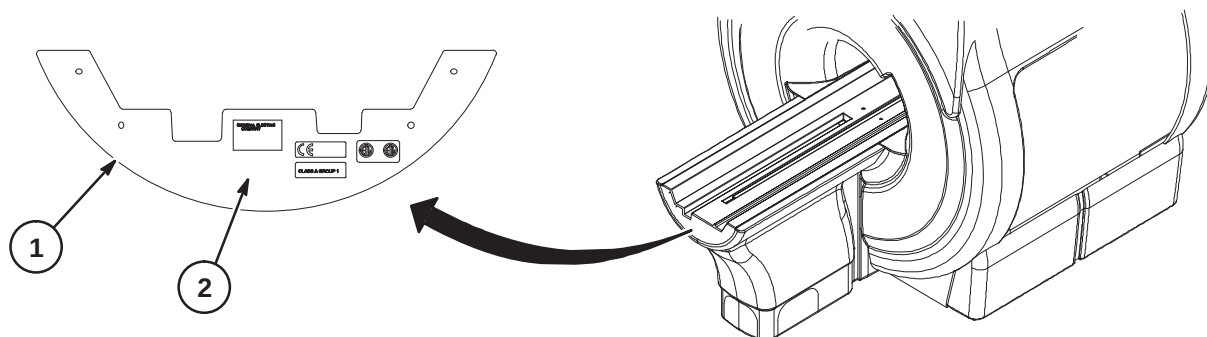
Note: Pedestal Covers may require an opening to be cut at appropriate location at bottom of cover(s) to allow routing of cables and hoses.

1. Hang new left and right Rear Pedestal Covers (5180369 & 5178111) on horizontal support bars.
2. If covers need adjustment, loosen the screws at each end of the horizontal support bar and reposition bar as necessary.
3. Secure left and right covers by fastening two M6 x 10mm Pan Head screws as shown below.



4-5-14 ATTACH REAR BRIDGE COVER AND RATING PLATES

1. If not already installed, attach Rear Bridge cover previously removed.
2. Attach all applicable rating plates.



SECTION 5 – COMPLETION

5-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 5-2, 5-3, and 5-4.

5-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 5-3. Recycling Operation will route them to appropriate location.

5-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-6. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

5-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Dock Patient Transport in position at front of bore.
3. Record and enter applicable data into applicable site configuration files and records.
4. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

5. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
6. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
7. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

5-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
4. On the PDU:
 - A. Move the mMain Input Circuit Breaker operating handle to OFF position (if in tripped state).
 - B. Move to ON (vertical) position.
 - C. Press the EMO Reset button.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the GOC.